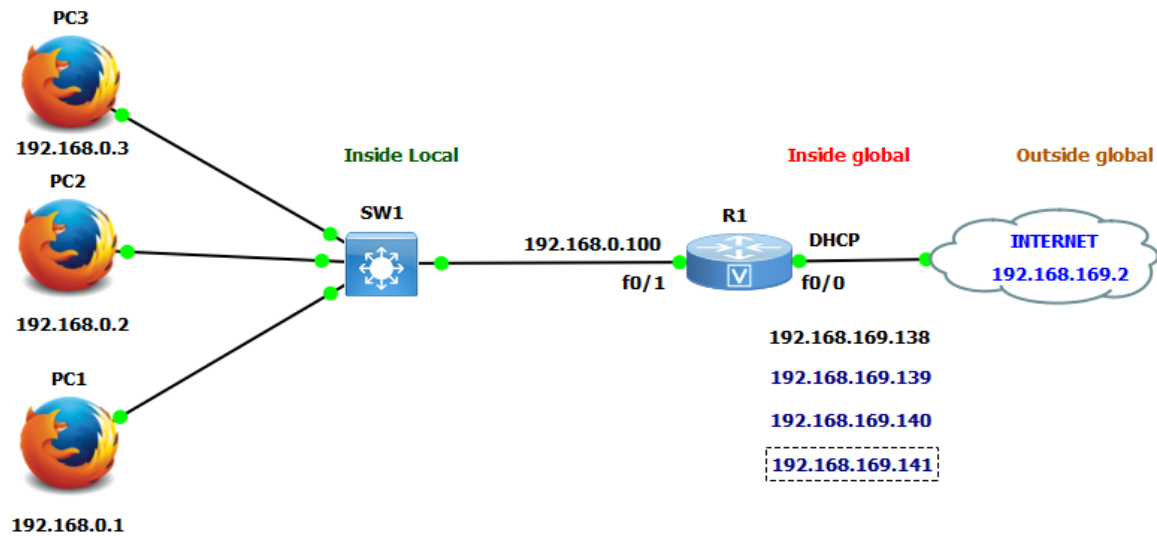


NAT/PAT Lab Time:



R1 Basic Configuration

R1(config)#interface f0/0	R1(config)#interface f0/1
R1(config-if)#ip address dhcp	R1(config-if)#ip add 192.168.0.100 255.255.255.0
R1(config-if)#no shutdown	R1(config-if)#no shutdown
R1(config)#ip name-server 8.8.8.8	R1(config)#ip domain-lookup

Static NAT Configuration on R1

R1(config)#ip nat inside source static 192.168.0.1 192.168.169.139
R1(config)#ip nat inside source static 192.168.0.3 192.168.169.141
R1(config)#ip nat inside source static 192.168.0.2 192.168.169.140
R1(config)#interface f0/0
R1(config-if)#ip nat outside
R1(config-if)#interface f0/1
R1(config-if)#ip nat inside

R1#show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	192.168.169.139	192.168.0.1	---	---
---	192.168.169.140	192.168.0.2	---	---
---	192.168.169.141	192.168.0.3	---	---

After send the traffic from 192.168.0.1, it is translated and send the traffic outside.

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
tcp 192.168.169.139:46834 192.168.0.1:46834 94.97.232.152:80 94.97.232.152:80
--- 192.168.169.139      192.168.0.1      ---              ---
tcp 192.168.169.140:41018 192.168.0.2:41018 94.97.232.152:80 94.97.232.152:80
```

Dynamic NAT Configuration on R1

```
R1(config)#access-list 1 permit 192.168.0.0 0.0.0.255
```

```
R1(config)#ip nat pool mypool 192.168.169.139 192.168.169.140 netmask 255.255.255.0
```

```
R1(config)#ip nat inside source list 1 pool mypool
```

```
R1(config)#interface f0/0
```

```
R1(config-if)#ip nat outside
```

```
R1(config-if)#interface f0/1
```

```
R1(config-if)#ip nat inside
```

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
tcp 192.168.169.140:39648 192.168.0.1:39648 93.184.220.29:80 93.184.220.29:80
tcp 192.168.169.140:46812 192.168.0.1:46812 13.249.11.44:443 13.249.11.44:443
tcp 192.168.169.140:54766 192.168.0.1:54766 50.112.78.76:443 50.112.78.76:443
tcp 192.168.169.140:59112 192.168.0.1:59112 13.249.11.17:443 13.249.11.17:443
--- 192.168.169.140      192.168.0.1      ---              ---
tcp 192.168.169.139:36886 192.168.0.3:36886 13.249.11.44:443 13.249.11.44:443
tcp 192.168.169.139:49660 192.168.0.3:49660 157.240.1.35:443 157.240.1.35:443
```

PAT Configuration on R1

```
R1(config)#access-list 1 permit 192.168.0.0 0.0.0.255
```

```
R1(config)#ip nat pool mypool 192.168.169.139 192.168.169.139 netmask 255.255.255.0
```

```
R1(config)#ip nat inside source list 1 pool mypool overload
```

```
R1(config)#interface f0/0
```

```
R1(config-if)#ip nat outside
```

```
R1(config-if)#interface f0/1
```

```
R1(config-if)#ip nat inside
```

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
tcp 192.168.169.139 44240 192.168.0.1 44240 52.88.19.211:443 52.88.19.211:443
tcp 192.168.169.139 33424 192.168.0.2 33424 93.184.220.29:80 93.184.220.29:80
tcp 192.168.169.139 33426 192.168.0.2 33426 93.184.220.29:80 93.184.220.29:80
tcp 192.168.169.139 33516 192.168.0.2 33516 93.184.220.29:80 93.184.220.29:80
tcp 192.168.169.139 37276 192.168.0.2 37276 54.187.176.55:443 54.187.176.55:443
tcp 192.168.169.139 50932 192.168.0.2 50932 94.97.232.147:80 94.97.232.147:80
tcp 192.168.169.139 60004 192.168.0.2 60004 35.163.69.146:443 35.163.69.146:443
tcp 192.168.169.139 60006 192.168.0.2 60006 35.163.69.146:443 35.163.69.146:443
tcp 192.168.169.139 60008 192.168.0.2 60008 35.163.69.146:443 35.163.69.146:443
tcp 192.168.169.139 60010 192.168.0.2 60010 35.163.69.146:443 35.163.69.146:443
tcp 192.168.169.139 60012 192.168.0.2 60012 35.163.69.146:443 35.163.69.146:443
tcp 192.168.169.139 35792 192.168.0.3 35792 34.211.202.13:443 34.211.202.13:443
```